EEE-101 - Control Systems Engineering

Tutorial - 3

1. Find the transfer function $\dfrac{X_1(s)}{F(s)}$

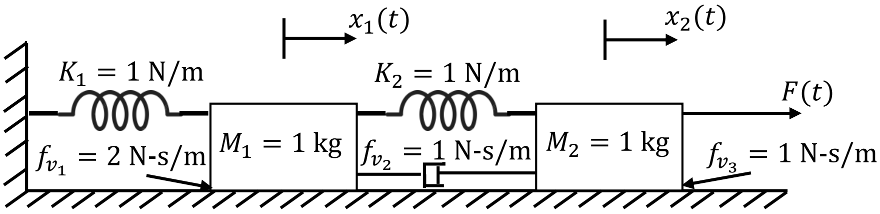

Figure 1

2. For the system shown below, find the transfer function $\dfrac{\theta_2(s)}{T(s)}$

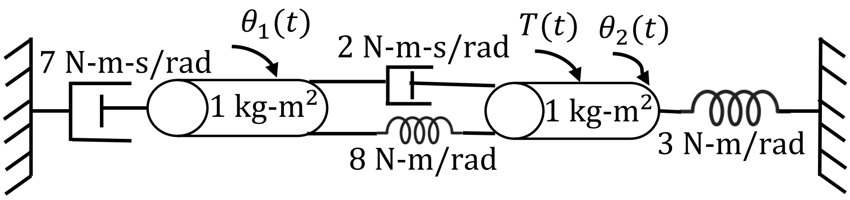

Figure 2

3. Given the electrical network in the below figure, find a state-space representation of the system dynamics considering $v_i(t)$ as the input and $v_o(t)$ as the output.

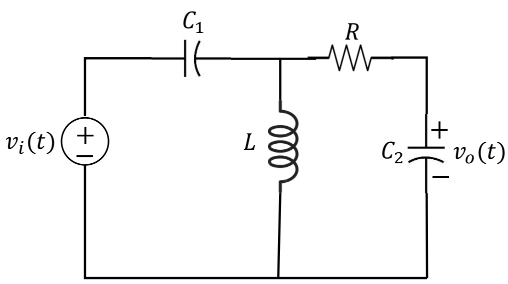

Figure 3

4. Find a state-space model of the system described by the given equation where $u(t)$ is the input and $y(t)$ is the output.

$$\frac{\mathrm{d}^3y(t)}{\mathrm{d}t^3} + 4\frac{\mathrm{d}^2y(t)}{\mathrm{d}t^2} + 6\frac{\mathrm{d}y(t)}{\mathrm{d}t} + 8y(t) = 10u(t).$$

5. The block diagram of a system with input u and outputs y_1 and y_2 is shown below. Find a state space model of the system.

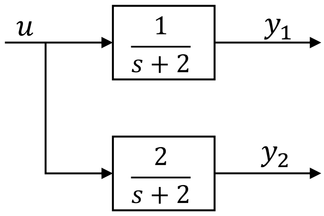

Figure 4

6. Consider the following state-space description of an LTI MIMO system.

$$\begin{bmatrix}\dot{x}_1\\ \dot{x}_2\end{bmatrix} = \begin{bmatrix}-1 & -1\\ 6.5 & 0\end{bmatrix}\begin{bmatrix}x_1\\ x_2\end{bmatrix} + \begin{bmatrix}1 & 1\\ 1 & 0\end{bmatrix}\begin{bmatrix}u_1\\ u_2\end{bmatrix},$$
$$\begin{bmatrix}y_1\\ y_2\end{bmatrix} = \begin{bmatrix}1 & 0\\ 0 & 1\end{bmatrix}\begin{bmatrix}x_1\\ x_2\end{bmatrix} + \begin{bmatrix}0 & 0\\ 0 & 0\end{bmatrix}\begin{bmatrix}u_1\\ u_2\end{bmatrix}.$$

Find its transfer function.

7. A system is represented by the following state-space model

$$\dot{x}(t) = \begin{bmatrix}0 & 1\\ 0 & -3\end{bmatrix} x(t) + \begin{bmatrix}1 & 1\\ 1 & 0\end{bmatrix} u(t),$$

$$y(t) = \begin{bmatrix}1 & 2\end{bmatrix} x(t)\cdot$$

Find the expression of $y(t)$, considering initial condition $x(0) = \begin{bmatrix}-1 & 3\end{bmatrix}^\top$ and the input $u(t)$ as a unit step.

8. A state-space representation of an LTI system is given as

$$\dot{x} = Ax + Bu, \quad x(0) := x_0 \in \mathbb{R}^n,$$
$$y = Cx + Du.$$

A new set of state variables (z) is chosen having a relation with the previous state vector (x) as $x = Tz$, where T is any $n \times n$ nonsingular matrix. The new state-space representation of the system becomes

$$\dot{z} = \bar{A}z + \bar{B}z, \quad z(0) := z_0,$$
$$y = \bar{C}z + \bar{D}u.$$

Show that the state transition matrices are related as $e^{\bar{A}t} = T^{-1}e^{At}T$.

9. An unforced LTI system is represented by

$$\dot{x}(t) = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -6 & -11 & -6 \end{bmatrix} x(t), \qquad x_0 = \begin{bmatrix} 1 \\ -3 \\ 4 \end{bmatrix}.$$

With a new choice of state variables (z), the system equation takes the form as

$$\dot{z}(t) = \bar{A}z(t), \quad \text{where } \bar{A} = \begin{bmatrix} -1 & 0 & 0 \\ 0 & -2 & 0 \\ 0 & 0 & -3 \end{bmatrix}.$$

Find the expression of $x(t)$ by evaluating $e^{\bar{A}t}$ (and by using the result of Q.6).